A Unified Performance and Parallelism Model for LLM Inference

Modeling Memory, FLOPs, and Collectives for Efficient Transformer Inference at Scale

Author: Yue Lu
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0. Introduction and Notation

This document presents a unified performance model for large-scale LLM inference on multi-GPU/NPU clusters. The focus is **autoregressive decoding**, where performance depends on:

- model dimensions $(H, n_q, n_{kv}, L, I_{\text{dense}}, I_{\text{moe}})$,
- parallelism configuration (DP, PP, EP, TP, SP),
- compute and memory efficiency (FlashAttention),
- KV-cache layout and size,
- interconnect topology and collective bandwidth.

0.1 Symbol Notation

Our symbol notation used in this documentation is aligned with:

- Megatron-LM (Shoeybi et al., 2019)
- DeepSpeed-MoE (Rajbhandari et al., 2022)
- FlashAttention (Dao et al., 2022, 2023)
- NVIDIA CUTLASS / cuBLAS GEMM dataflow
- Recent MoE and long-context LLM deployments

Parallelism Dimensions

- DP — Data Parallelism
  Full model replicated across devices. Each DP replica handles disjoint input batches. Mostly relevant for throughput scaling across independent requests.

- PP — Pipeline Parallelism
  Layers split into pipeline stages. Let total layers be L and pipeline degree be PP, then each stage holds $L_{\text{stage}} = \frac{L}{PP}$

- TP — Tensor Parallelism
  Splits large matrix multiplies across devices. Used to shard attention Q/K/V/O projections and FFN/MoE matrices. Commonly “column” or “row” parallel as in Megatron-LM.

- EP — Expert Parallelism
  For MoE models: experts are partitioned across devices. Tokens are routed via all-to-all within an EP group, so each device hosts a subset of experts.

- SP — Sequence Parallelism
  In this document, SP means **ring-attention-style KV sharding** for inference: the sequence dimension is partitioned across devices for KV storage, while each rank still computes attention for all queries via ring communication.

Model Dimensions

- L — Number of transformer layers.

- V — Vocabulary size (number of tokens in the tokenizer).

- H — Hidden size (model dimension). Hidden-state dimension H applies to embeddings, LM head, FFN blocks, and attention projections.
- n_q — Number of query heads.
- d_{head} — Head dimension, typically H/n_q.
- n_{kv} — Number of KV heads (in GQA, $n_{kv} < n_q$).
- $H_{kv} = n_{kv} d_{\text{head}}$ — Total KV projection dimension. KV dimension H_{kv} applies specifically to K/V projections and cache storage.
- I — Unified FFN intermediate dimension
  - $I = I_{\text{dense}}$ for dense layers
  - $I = I_{\text{moe}}$ for MoE layers
- I_{eff} — Unified FFN intermediate dimension for FLOPs calculation
  - $I_{\text{eff}} = I_{\text{dense}}$ for dense layers
  - $I_{\text{eff}} = kI_{\text{moe}}$ for MoE layers
- N_{exp} — Number of experts per MoE layer
- N_{eff} — Unified number of experts per MoE layer for FLOPs calculation
  - $N_{\text{eff}} = 0$ for dense layers
  - $N_{\text{eff}} = N_{\text{exp}}$ for MoE layers
- k — Number of experts selected per token (top-k routing in MoE)

Sequence / Batch and Precision

- S — Sequence length (tokens in context)
- B — Batch size (number of sequences)
  For decoding, B does *not* affect memory footprint or per-token FLOPs. It only appears in prefill (Section 3.6 and Section 6.4).
  Thus, throughout this document, B is included only where prefill behavior is analyzed.
- b — Bytes per element (e.g., bf16 ⇒ $b = 2$, fp8 ⇒ $b = 1$)

Static and Dynamic Memory

Parameter size

- P_{attn} — Attention parameter size
- P_{FFN} — Unified FFN/MoE parameter size
- P_{emb} — Embedding parameter size
- P_{lm} — LM head (0 if tied) parameter size

Memory capacity quantities (how many bytes are **stored** in HBM)

- $M_{\theta,\text{device}}$ — Parameter memory on this device
- $M_{\text{KV,device}}$ — KV cache storage for keys + values
- $M_{\text{act,device}}$ — Activation working memory on this device (per token during decoding)
- M_{HBM} — Available HBM capacity per device

Memory traffic quantities (how many bytes **move** between HBM and compute per token)

- $T_{\theta,\text{device}}$ — Parameter traffic (weights read per token)
- $T_{\text{KV,device}}$ — KV traffic (KV read + write per new token)
- $T_{\text{act,device}}$ — Activation traffic (intermediate reads/writes)
- $T_{\text{token,device}}$ — Total per-token traffic on this device
- $T^{eff}_{\text{token,device}}$ — Effective traffic after FlashAttention-style optimizations

We also define:

- c_{act} — Empirical number of unavoidable hidden-state I/O per layer that remains even after FlashAttention-style optimizations.

Device Compute and Bandwidth

- R_{GPU} — Effective compute throughput of the device (FLOPs/s).
- γ_{FA} — FlashAttention factor that encodes **memory efficiency gains** from FlashAttention-style kernels:
  - reduces memory traffic for Q/K/V and attention intermediates,
  - increases effective memory bandwidth utilization.
- $B_{\text{eff,mem}}$ — Effective HBM bandwidth (bytes/s).

Networking

- $\alpha_{TP}, \alpha_{EP}, \alpha_{SP}, \alpha_{PP}$ — Startup latency constants for TP, EP, SP, and PP collectives, respectively (modeled in an α–B style latency + bandwidth formulation).
- $B_{\text{eff,TP}}$, $B_{\text{eff,EP}}$, $B_{\text{eff,SP}}$, $B_{\text{eff,PP}}$ — Effective interconnect bandwidths for TP, EP, SP, and PP communication.
- n_{TP} — Number of TP collective iterations per-layer per-token
- n_{EP} — Number of EP collective iterations per-layer per-token

Model FLOPs and Compute Quantities

FLOPs (floating point operations) per token and per layer:

- F_Q, F_K, F_V, F_O — FLOPs for the Q, K, V, and output projections.
- F_{proj} — Combined Q/K/V/O projection FLOPs.
- F_{score} — FLOPs to compute attention scores ($QK^\top$).
- F_{value} — FLOPs to apply attention weights to values.
- $F_{\text{attn,KV}}$ — FLOPs for the attention score + value application terms.
- F_{attn} — Total attention FLOPs per layer (projections + attention score/value application).

FFN and MoE FLOPs:

- $F_{\text{ffn,dense}}$ — Dense FFN FLOPs per layer.
- F_{router} — Router FLOPs per token for MoE.
- F_{expert} — FLOPs per expert MLP (for one expert and one token).
- $F_{\text{ffn,moe}}$ — MoE FFN FLOPs per layer.
- F_{ffn} — Unified FFN FLOPs (dense or MoE), used throughout Section 5.

Layer and token FLOPs:

- F_{norm} — FLOPs for normalization (LayerNorm/RMSNorm), typically $c_{\text{norm}}H$.

- c_{norm} — Constant capturing the norm FLOP coefficient (e.g., 5–10).
- F_{layer} — Total FLOPs per layer (attention + FFN + norm).
- $F_{\text{layer,device}}$ — FLOPs per layer per device after TP/EP sharding.
- $F_{\text{token,device}}$ — Total FLOPs per generated token on this device (decoding).
- F_{prefill} — Total FLOPs for prefill (processing the full input sequence before decoding).

Time, Throughput, and Roofline Quantities

Local (on-device) & communication timing:

- t_{compute} — Per-token compute time, $F_{\text{token,device}}/R_{\text{GPU}}$.
- t_{mem} — Per-token memory time, $T^{eff}_{\text{token,device}}/B_{\text{eff,mem}}$.
- t_{local} — Local per-token roofline time:
- t_{TP}, t_{EP}, t_{SP}, t_{PP} — Communication time per token for TP/EP/SP/PP.
- t_{comm} — Combined communication work per token.
- $t_{\text{prefill,local}}$ — Prefill compute + memory time.
- $t_{\text{prefill,comm}}$ — Prefill communication time.
- t_{startup} — Kernel warmup and graph-capture overhead.
- t_{token} — Effective per-token time, accounting for compute, memory, and communication.
- ρ — Empirical overlap factor for communication in the practical model (e.g., $\rho \in [0.3, 1]$), used in approximations such as $t_{\text{token}} \approx \max(t_{\text{local}},\ \rho\, t_{\text{comm}})$

Throughput & end-to-end latency:

- TPS_{single} — Per-DP replica throughput: token per second
- $TTPS$ — Global throughput across DP replicas: total token per second TTPS is the correct metric for end-to-end system throughput when multiple DP replicas serve independent requests.
- $TTFT$ — Time to first token: the amount of time an LLM takes to generate the first token in its response after receiving an input or prompt.

0.2 Parallelism Architecture

We adopt a fixed nesting order for all parallelism dimensions:

$$\boxed{\text{DP} \ \rightarrow\ \text{PP} \ \rightarrow\ \text{EP} \ \rightarrow\ \text{TP} \ \rightarrow\ \text{SP}}$$

This order reflects how model state is partitioned and reused during inference. Each level depends on all outer levels having already determined the weight placement, token routing, or tensor partitioning.

Summary of Nesting Rationale

Level	What it partitions	Why this ordering is required
DP	Entire model replica	Must wrap all state; inner groups cannot cross DP boundaries.
PP	Layers	Layer ownership must be decided before experts/tensor shards are assigned.
EP	Experts	Expert placement must be fixed before tensor sharding splits expert matrices.
TP	Weight matrices	TP defines weight shards used identically across all SP ranks.

SP	KV cache sequence dimension	KV is activation state only; must be sharded after all weight placement.

DP is the outermost level (replicated model weights)

Why:

- DP creates fully independent model replicas for throughput.
- No weight partitioning happens inside DP groups.
- All deeper parallelism dimensions (PP, EP, TP, SP) apply **within** each DP replica.

Therefore:

$$\text{DP is always outermost}$$

PP is inside DP (layers assigned before experts/tensor sharding)

PP splits *layers* of the model across devices.

Why PP must come before EP, TP, SP:

- PP determines **which layers live on which devices**.
- Only after PP is fixed can we partition:
  - Experts (EP) across devices
  - Tensor dimensions (TP) inside each block
  - KV sequence partitions (SP) inside each attention layer
- PP stages own their local KV cache and local weights.

Therefore:

$$\text{DP} \to \text{PP}$$

EP is inside PP (expert groups belong to specific layers)

EP distributes MoE experts **within the layers assigned by PP**.

Why EP must come before TP and SP:

- Expert weights must be assigned to EP ranks resolved **before** tensor-parallel shards apply.
- TP cannot shard expert weights until expert placement is set by EP.
- SP does not affect expert routing; EP must be outer.

In addition, for each MoE layer with N_{exp} experts, the expert-parallel degree must satisfy

$$EP \leq N_{\text{exp}},$$

so that each EP rank can host at least one expert. In practice, EP is usually chosen to divide N_{exp} (e.g., $EP \mid N_{\text{exp}}$), but the only hard requirement in this model is $EP \leq N_{\text{exp}}$.

Therefore:

$$\text{DP} \to \text{PP} \to \text{EP}$$

TP is inside EP (tensor sharding within a defined expert/layer partition)

TP splits matrices *within each expert or dense block*.

Why TP must come before SP:

- TP shards Q/K/V/O projections, MLP layers, and expert MLPs.
- After TP, each rank holds a **fraction of H or H_{kv}**.
- SP requires all SP ranks to share identical TP-sharded weights.

Therefore:

$$\mathrm{DP} \to \mathrm{PP} \to \mathrm{EP} \to \mathrm{TP}$$

SP is innermost (KV sharding after all weights are fixed)

SP shards the **KV cache**, not model parameters.

Why SP must be last:

- SP only partitions activation state (KV), not weights.
- SP requires all TP ranks to already have consistent weight shards.
- Only after DP/PP/EP/TP are fixed can we shard the KV cache.

Therefore:

$$\mathrm{DP} \to \mathrm{PP} \to \mathrm{EP} \to \mathrm{TP} \to \mathrm{SP}$$

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1. Memory Footprint

This section defines parameter sizes and memory footprint for a given set of model parameters. The memory footprint include those from model weights, per-token activation/working memories, and KV cache. We avoid model-wide parameter aggregation here and instead focus on **per-layer** quantities, because pipeline-parallel stages own disjoint sets of layers.

All parameter definitions assume stored precision of b bytes per element (e.g., bf16 = 2 bytes).

1.0 Parameter Definitions (P vs W)

For any weight matrix W, the parameter count is

$$P(W) = \text{number of elements in } W$$

Total stored parameter memory is

$$M_\theta = P \cdot b$$

1.1 Model Parameter Memory

Embedding and LM Head Parameters

Modern LLM architectures (GPT-3/4, LLaMA families, PaLM, Qwen, DeepSeek, etc.) typically use embedding dimension $E = H$, so all internal projections operate on vectors in $\mathbb{R}^H$.

For the token embedding:

$$W_{\text{emb}} \in \mathbb{R}^{V \times H}, \quad P_{\text{emb}} = VH$$

If the LM head is tied with embeddings, i.e. $W_{lm} = W_{emb}^T$ so that W_{emb} can be re-used:

$$P_{\text{lm}} = 0$$

If untied:

$$W_{\text{lm}} \in \mathbb{R}^{H \times V}, \quad P_{\text{lm}} = VH$$

Attention Parameters

For hidden size H, head dimension d_{head}, and KV dimension $H_{kv} = n_{kv}\, d_{\text{head}}$:

- $W_Q \in \mathbb{R}^{H \times H}$
- $W_K \in \mathbb{R}^{H \times H_{kv}}$
- $W_V \in \mathbb{R}^{H \times H_{kv}}$
- $W_O \in \mathbb{R}^{H_{kv} \times H}$

Parameter counts:

$$P_Q = H^2, \qquad P_K = HH_{kv}, \qquad P_V = HH_{kv}, \qquad P_O = H_{kv}H$$

We define attention parameters:

$$P_{\text{attn}} = P_Q + P_K + P_V + P_O$$

Unified FFN Parameters (Dense or MoE)

Each transformer layer contains an FFN module. In modern LLM architectures, this FFN is almost always implemented as a gated MLP (GeLU/SiLU/GLU/SwiGLU–style) with **three** linear projections:

- a gate projection,
- an "up" projection (expansion),
- a "down" projection (contraction).

For a hidden size H and FFN intermediate dimension I, this yields an FFN parameter count

$$P_{\text{FFN}} = 3HIN_{\text{exp}}.$$

Here N_{exp} denotes the number of experts **per layer**:

- **For a dense MLP model:** $I = I_{\text{dense}},\ N_{\text{exp}} = 1.$
- **For a MoE model:** $I = I_{\text{moe}}$, with $N_{\text{exp}} > 1.$

These two model cases are mutually exclusive **per layer**.

LayerNorm parameters

LayerNorm or RMSNorm contain $\mathcal{O}(H)$ parameters (scale and optional bias). These are negligible compared to attention and FFN weights and are omitted in scaling formulas.

1.2 Activation Memory (Per-token Working Memory)

During autoregressive decoding, the model processes **one token at a time**. As a result, the only activations that need to be stored in memory are the **temporary, layer-local working buffers** used in the forward pass of the *current token*.

These activations are **not reused** across layers or across tokens and therefore are:

- **not dependent on sequence length** S,
- **not dependent on batch size** B,
- **not EP- or TP-sharded**,
- and **extremely small** relative to model parameter memory (Section 1.1) and KV cache memory (Section 1.3).

Below we account for all activation tensors in a model layer that must be alive **concurrently** for one decoding token.

Q, K, V projections

For the current hidden state $h \in \mathbb{R}^H$, the layer computes:

- $Q \in \mathbb{R}^H$
- $K \in \mathbb{R}^{H_{kv}}$
- $V \in \mathbb{R}^{H_{kv}}$

This contributes:

$$H + 2H_{kv}$$

Attention score accumulation buffer (FlashAttention-like kernels)

Attention score computation normally requires a temporary buffer.
FlashAttention-style fused kernels avoid storing full S-length score vectors and instead use a **single internal workspace** of size H during streaming softmax.

This adds:

$$+H$$

Attention output buffer

After applying attention weights to V and combining across heads, we form the attention output $O_{\text{attn}} \in \mathbb{R}^H$.

This output must exist before the output projection is applied, contributing:

$$+H$$

FFN working buffer

Following attention and normalization, the FFN block needs at least one temporary buffer of size H to hold either the FFN input or the FFN output before residual addition.
Even with kernel fusion, this buffer cannot always overlap with the attention intermediates.

This adds:

$$+H$$

Summing all simultaneously required buffers:

$$P_{\text{act}} = 4H + 2H_{kv}$$

In bytes:

$$M_{\text{act,layer}} = (4H + 2H_{kv}) \cdot b$$

This footprint is **small** compared to parameter memory and KV cache, which is why activation memory does not limit decoding capacity under typical inference workloads.

1.3 KV Cache Memory

Section 1.1 described the memory footprint of model parameters (static), and Section 1.2 covered the activation memory required during decoding (per-token, dynamic). This section describes the **KV cache**, which is a *runtime* structure generated during the **pre-fill phase**, when the model processes the entire input sequence of length S.

During pre-fill, each attention layer produces:

- one key vector of dimension H_{kv},
- one value vector of dimension H_{kv},

for each input token. Because decoding adds only one new token at a time, the vast majority of KV memory comes from **pre-fill**, not decoding.

For a single attention layer, the KV cache consists of:

- Keys: $K \in \mathbb{R}^{S \times H_{kv}}$
- Values: $V \in \mathbb{R}^{S \times H_{kv}}$

Thus, the total KV elements for one layer are:

$$P_{KV,layer} = S \cdot (2H_{kv}) = 2SH_{kv}$$

In bytes:

$$M_{\text{KV,layer}} = 2SH_{kv} \cdot b$$

This is **static** once pre-fill is complete; decoding contributes only an additional $2H_{kv}b$ per generated token, which is negligible relative to the full cache.

1.4 Per-device Memory Footprint After Parallelism Sharding

So far we've completed the all the memory footprint estimation for a model layer. When we introduce different parallelism schemes, some of these memories would be sharded by one or more of these parallelism dimensions, resulting in a somewhat complicated memory aggregateion per device. We now describe how these parameters are distributed across devices under PP/EP/TP/SP, and then derive simple modeling approximations.

Per-device Parameter Memory (TP, EP, PP)

Each transformer layer has two parameter groups:

- **Attention parameters** P_{attn}, that are sharded by **tensor parallelism (TP)** only.
- **FFN parameters** P_{FFN} are sharded by **TP and EP**. Dense FFN layers are treated as a special case with $EP = 1$, so the same formula applies to both dense and MoE FFNs.

For a *single layer* on a device, the stored parameter memory is

$$M_{\theta,\text{layer}} = \frac{P_{\text{attn}}\, b}{TP} + \frac{P_{\text{FFN}}\, b}{TP \cdot EP}$$

Pipeline parallelism (PP) assigns **disjoint sets of layers** to different stages.
Let L_s be the set of layers that live on PP stage s, and let $M_{\theta,\text{layer},\ell}$ be the per-layer memory from the expression above.

Excluding embeddings and LM head, the parameter memory per device on PP stage s is

$$M_{\theta,\text{layers}}^{(s)} = \sum_{\ell \in L_s} M_{\theta,\text{layer},\ell}$$

Embeddings and LM head appear only on two stages:

- **Intermediate PP stages** (no embedding / LM head):

$$M_{\theta,\text{device}}^{(\text{mid})} = M_{\theta,\text{layers}}^{(\text{mid})}$$

- **First PP stage** (with token embedding):

$$M_{\theta,\text{device}}^{(1)} = M_{\theta,\text{layers}}^{(\text{mid})} + \frac{P_{\text{emb}}\, b}{TP}$$

- **Final PP stage** (with LM head):

$$M_{\theta,\text{device}}^{(\text{PP})} = M_{\theta,\text{layers}}^{(\text{mid})} + \frac{P_{\text{lm}}\, b}{TP}$$

If each intermediate PP stage holds approximately L/PP layers of similar size, and we use representative per-layer values P_{attn} and P_{FFN}, then

$$M_{\theta,\text{device}}^{(\text{mid})} = \frac{L}{PP} M_{\theta,\text{layer}} = \frac{L}{PP} \left(\frac{H^2 + 3HH_{kv}}{TP} + \frac{2HIN_{\text{exp}}}{TP \cdot EP} \right) b$$

For capacity planning we use a worst-case PP stage budget, adding one $\frac{VH}{TP} b$ term to account for embedding/LM weights residing on boundary stages. Intermediate stages are slightly smaller. Therefore:

$$M_{\theta,\text{device}} = \frac{L}{PP} \left(\frac{H^2 + 3HH_{kv}}{TP} + \frac{3HIN_{\text{exp}}}{TP \cdot EP} \right) b + \frac{VH}{TP} b$$

For a **dense MLP model**: $I = I_{\text{dense}}$, and $N_{\text{exp}} = EP = 1$.

For a **MoE model**: $I = I_{\text{moe}}$, with $N_{\text{exp}} > 1$ and $EP \geq 1$.

Per-device Activation Memory

The per-layer working activation footprint for one decoding token is: $4H + 2H_{kv}$.

This footprint must exist for every layer on the PP stage. Since PP assigns whole layers to devices, each device hosts exactly L/PP layers. Therefore, the per-device activation memory during decoding is:

$$M_{\text{act,device}} = \frac{L}{PP} (4H + 2H_{kv})\, b$$

Per-device KV Cache Memory

Only attention layers produce KV cache.
The KV cache is sharded only across:

- **TP** — splits the channel dimension H_{kv},
- **SP** — splits the sequence dimension S.

EP and DP do not modify KV layout; PP only affects how many layers are assigned to a stage.

Therefore, the **per-device** KV memory footprint is:

$$M_{\text{KV,device}} = \frac{L}{PP} \frac{M_{\text{KV,layer}}}{TP \cdot SP} = \frac{L}{PP} \frac{(2SH_{kv})b}{TP \cdot SP}$$

which:

- For long-context inference (e.g., $S \in [16\text{k}, 128\text{k}]$), $S(2H_{kv})b$ is large enough that KV can exceed parameter memory unless aggressively reduced through TP and SP.
- Each decoded token adds only $2H_{kv}b$, which is negligible compared to the pre-fill KV footprint.

Total Per-device Static Memory Footprint

Summing all the memory footprint we dervive from section 1.1 - 1.4 together, we can therefore get the "minimum required" memory size for the device to host the model under a particular PP/EP/TP/SP partition.

$$M_{\text{device}}^{\text{total}} = M_{\theta,\text{device}} + M_{\text{act,device}} + M_{\text{KV,device}}$$

$$\boxed{M_{\text{device}}^{\text{total}} = \frac{L}{PP}\left(\frac{H^2 + 3HH_{kv}}{TP} + \frac{2HIN_{\text{exp}}}{TP \cdot EP} + (4H + 2H_{kv}) + \frac{2SH_{kv}}{TP \cdot SP}\right)b + \frac{VH}{TP}b}$$

For a dense model: $I = I_{\text{dense}},\ N_{\text{exp}} = EP = 1$

And for a MoE model: $I = I_{\text{moe}}$

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2. Memory Traffic During Decoding

Section 1 quantified the *static* memory footprint of the model — how many bytes of parameters, KV cache, and activations must **fit** in device HBM.

This section instead focuses on **memory traffic per generated token**, i.e., the bytes that must flow between HBM and compute cores *during decoding*. This traffic directly determines the memory-bound component of decoding performance (Section 4's roofline model).

We reuse the same parameter partitions from Section 1 to ensure consistency.

2.1 Model Parameter Traffic $T_{\theta,\text{device}}$

Following Section 1, we use:

- P_{attn}: Q/K/V/O projection parameters
- P_{FFN}: dense FFN (or non-expert MoE core) parameters

We also have:

- P_{emb}: embedding matrix
- P_{lm}: LM head

but **as in the intermediate-stage memory model of Section 1.4**, we drop P_{emb} and P_{lm} when modeling steady-state decoding, since they are small relative to attention and FFN parameters and typically reside only on boundary PP stages.

FlashAttention and the role of γ_{FA}

FlashAttention reduces DRAM traffic for attention weights by aggressively tiling Q/K/V/O blocks into SRAM and reusing them across the fused attention kernels.

The impact is well documented:

- **FlashAttention-1** (Dao et al. 2022): ~2× reduction
- **FlashAttention-2** (Dao et al. 2023): 2.5×–4× reduction
- **FlashAttention-3** (Dao et al. 2024): 3×–5× reduction
- **xFormers / FasterTransformer / TensorRT-LLM**: typically 2×–3.5× reduction

Thus a practical range is:

$$\gamma_{FA} \approx 2\text{–}4 \quad (\text{typical}), \qquad \gamma_{FA} \approx 4\text{–}5 \quad (\text{FlashAttention-3}).$$

This factor applies **only to attention-parameter traffic**.

Attention parameter traffic

Because P_{attn} is defined **per layer**, a PP stage with $L_{\text{stage}} = L/PP$ layers has $L_{\text{stage}} P_{\text{attn}}$ attention parameters in total (before TP sharding).

With FlashAttention reducing attention-parameter traffic by γ_{FA}, the per-token attention parameter traffic per device in this PP stage is

$$T_{\theta,\text{attn}} \approx \frac{L}{PP} \cdot \frac{P_{\text{attn}}\, b}{TP \cdot \gamma_{FA}}$$

FFN parameter traffic

Modern FFN/MLP implementations (e.g., GLU/SwiGLU-based MLPs in LLaMA, Qwen, PaLM) use **three** linear projections (gate, up, down), giving the true parameter count

$$P_{\text{FFN}} = 3HIN_{\text{exp}}.$$

However, optimized inference kernels such as **FlashMLP**, TensorRT-LLM fused MLP, and Megatron-Core grouped-GEMM fuse the **gate** and **up** projections into a single memory stream. As a result, the *effective* memory traffic of the FFN behaves closer to a **2-projection** design.

To model this we introduce an empirical reduction factor

$$\gamma_{\text{FMLP}} \approx 1.5,$$

so that the **effective FFN parameter-traffic contribution** per-token per-device in this PP stage is becomes

$$T_{\theta,\text{FFN}} = \frac{L}{PP}\ \frac{P_{\text{FFN}}}{TP \cdot EP \cdot \gamma_{\text{FMLP}}}\ b$$

This mirrors the use of γ_{FA} for attention and highlights that in practice, FFN parameter traffic is **significantly lower** than its raw 3-projection parameter count would suggest.

Final approximate parameter-traffic expression

Dropping P_{emb} and P_{lm} (as in the intermediate-stage memory model of Section 2), we obtain

$$T_{\theta,\text{device}} \approx \frac{L}{PP}\left(\frac{P_{\text{attn}}}{TP \cdot \gamma_{FA}} + \frac{P_{\text{FFN}}}{TP \cdot EP \cdot \gamma_{\text{FMLP}}}\right) b = \frac{L}{PP}\ \left(\frac{H^2 + 3HH_{kv}}{TP \cdot \gamma_{FA}}\ +\ \frac{3HIN_{\text{exp}}}{TP \cdot EP \cdot \gamma_{\text{FMLP}}}\right) b$$

For a dense MLP model: $I = I_{\text{dense}},\ N_{\text{exp}} = EP = 1$

And for a MoE model: $I = I_{\text{moe}}$

2.2 Activation Traffic $T_{\text{act,device}}$

Section 1 showed that the per-layer activation footprint for a single decoding token is small. These H-dimensional vectors must still be loaded/stored at HBM during attention and FFN/MoE processing. Unlike KV cache, activation traffic does **not** scale with the sequence length S.

Empirical kernel traces from FlashAttention-2/3 (Dao et al. 2022–2024), xFormers (Meta 2023), TensorRT-LLM, and FasterTransformer consistently show:

- ~5–12 reads/writes of an H-vector per layer per decoding step.

Therefore we approximate:

$$T_{\text{act,layer}} \approx c_{\text{act}}\, H\, b, \qquad c_{\text{act}} \approx 8\text{–}12.$$

Relationship between c_{act} and γ_{FA}

It is important to distinguish:

- γ_{FA} — reduces **attention parameter traffic** by tiling and fusing Q/K/V/O computations to minimize DRAM reads of weights and attention intermediates.
- c_{act} — captures the **residual activation I/O that remains even after** the FlashAttention optimization (e.g., reading/writing the hidden state, attention accumulators, FFN input/output buffers).

FlashAttention eliminates most of the attention *score-path* memory movement, but as noted in the FlashAttention papers and in Meta and NVIDIA kernel traces, it **cannot eliminate hidden-vector I/O**. These unavoidable loads/stores are precisely what c_{act} models. Thus:

- c_{act} **is already a "post-FlashAttention" empirical constant**,
- and we **do not** apply γ_{FA} again to $T_{\text{act,device}}$.

Final activation-traffic expression

For a PP stage containing L/PP layers:

$$T_{\text{act,device}} = \frac{L}{PP}\, c_{\text{act}}\, H\, b$$

As with activation *memory* in Section 4, this traffic is **not sharded** by TP, EP, or SP because:

- the hidden state $h \in \mathbb{R}^H$ must be replicated across TP ranks, and
- fused kernels require the full activation vector before any weight sharding applies.

Activation traffic is typically smaller than KV and parameter traffic but is included for completeness.

2.3 KV Cache Traffic $T_{\text{KV,device}}$

KV cache must be **fully read** for each new token:

- Keys: $S \cdot H_{kv}$
- Values: $S \cdot H_{kv}$
- Write new K/V: $2H_{kv}$

For large S, the write term is negligible, so the per-layer KV access is approximately:

$$T_{\text{KV,layer}} \approx 2SH_{kv}\, b$$

Consistent with Section 1, KV is sharded by **TP** (channel/head dimension) and **SP** (sequence dimension), but *not* by EP, DP, or PP. Thus each device only sees a $\frac{1}{TP \cdot SP}$ shard of this per-layer traffic:

$$T_{\text{KV,layer}} \approx \frac{2SH_{kv}\, b}{TP \cdot SP}$$

A PP stage holds L/PP attention layers, so the **per-device KV traffic per token** on this PP stage is:

$$T_{\text{KV,device}} \approx \frac{L}{PP} \cdot \frac{2SH_{kv}\, b}{TP \cdot SP}$$

Note that some attention kernel implementations (e.g., FA-2 streaming kernels) may read portions of K or V twice due to numerical scaling or alignment. This introduces a small multiplicative overhead (typically 1.1–1.3×).

For analytical modeling, we treat KV reads as exactly $2SH_{kv}b$ per layer, which matches the dominant asymptotic behavior.

FlashAttention **does not** reduce KV traffic: keys and values must always be read in rull (and new entries written) for every decoded token, regardless of kernel fusion or tiling strategy.

2.4 Total and Effective Traffic

Combining the expressions derived in Sections 2.1–2.3, the **effective** total per-token traffic is simply:

$$T^{eff}_{\text{token,device}} \approx T_{\theta,\text{device}} + T_{\text{act,device}} + T_{\text{KV,device}}.$$

Substituting everything yields the **final fully expanded expression**:

$$\boxed{T^{eff}_{\text{token,device}} \approx \frac{L}{PP}\left(\frac{H^2 + 3HH_{kv}}{TP \cdot \gamma_{FA}} + \frac{3HIN_{\text{exp}}}{TP \cdot EP \cdot \gamma_{FMLP}} + c_{\text{act}}\, H + \frac{2SH_{kv}}{TP \cdot SP}\right) b}$$

2.5 Static Memory Footprint vs. Memory Traffic (Important Distinction)

Sections 1 and 2 play different roles in the overall performance model:

Static Memory Footprint (Section 1)

Determines whether a (DP, PP, EP, TP, SP) configuration can *fit* on a device:

$$M^{\text{total}}_{\text{device}} = M_{\theta,\text{device}} + M_{\text{act,device}} + M_{\text{KV,device}} \leq M_{\text{HBM}}.$$

Using the fully expanded expression from Section 1:

$$M^{\text{total}}_{\text{device}} = \frac{L}{PP}\left(\frac{H^2 + 3HH_{kv}}{TP} + \frac{3HIN_{\text{exp}}}{TP \cdot EP} + (4H + 2H_{kv}) + \frac{2SH_{kv}}{TP \cdot SP}\right) b + \frac{VH}{TP} b.$$

This is a **capacity constraint**:

> *"Does the entire model and KV-cache fit in HBM?"*

Memory Traffic (Section 2)

Determines the *bandwidth-limited latency* per decoded token:

$$T^{eff}_{\text{token,device}} \approx \frac{L}{PP}\left(\frac{H^2 + 3HH_{kv}}{TP\gamma_{FA}} + \frac{3HIN_{\text{exp}}}{TP \cdot EP \cdot \gamma_{FMLP}} + c_{\text{act}} H + \frac{2SH_{kv}}{TP \cdot SP}\right) b.$$

This traffic produces a memory-bound latency:

$$t_{\text{mem}} = \frac{T^{eff}_{\text{token,device}}}{B_{\text{eff,mem}}}.$$

This is a **bandwidth constraint**:

> *"Given this sharding, how slow is HBM?"*

Why this distinction matters

- Section 1 tells us **which parallelism configurations are even viable** under HBM limits.
- Section 2 tells us **how fast decoding can proceed** for those viable configurations.
- Section 6 will combine:
  - static footprint (HBM capacity),
  - memory-bound latency t_{mem},
  - compute-bound latency t_{compute},
  - and communication latency t_{comm}.

Together they determine the **per-token throughput**, **pipeline bottleneck**, and **TTFT**.

---

3. Compute (FLOPs) per Token

During inference, the FLOPs required to generate a token depend on the transformer layer structure. We distinguish between:

- **Prefill FLOPs (GEMM-dominant):** $O(S^2)$ across the full input sequence
- **Decoding FLOPs (GEMV-dominant):** $O(S)$ for one additional token

This section focuses on **decoding FLOPs**, which determine TPS throughput. All FLOPs below represent **per-token**, **per-layer**, **decoding** FLOPs.

3.1 Q/K/V and Output Projections

Q, K, and V projections are vector–matrix multiplications of shapes:

- Q: $[1 \times H] \cdot [H \times H]$
- K: $[1 \times H] \cdot [H \times H_{kv}]$
- V: $[1 \times H] \cdot [H \times H_{kv}]$
- Output: $[1 \times H_{kv}] \cdot [H_{kv} \times H]$

Projection FLOPs

$$F_Q = 2H^2, \qquad F_K = 2HH_{kv}, \qquad F_V = 2HH_{kv}, \qquad F_O = 2HH_{kv}.$$

Where the factor 2 accounts for each multiply-accumulate pair in the standard GEMV convention.

Total

$$F_{\text{proj}} = 2H^2 + 6HH_{kv}$$

If $H_{kv} = H$, this reduces to $8H^2$.

3.2 Attention Scores and Value Application

During decoding, the newly generated token attends to all S cached tokens in the KV cache for this layer. Conceptually, for each layer we can treat the cached keys and values as:

- $K_{\text{cache}} \in \mathbb{R}^{S \times H_{kv}}$
- $V_{\text{cache}} \in \mathbb{R}^{S \times H_{kv}}$,

where $H_{kv} = n_{kv} d_{\text{head}}$ is the total KV projection dimension.

Scores (Q · Kᵀ)

For the new token, attention scores are computed by taking a dot product between its query representation and each of the S cached key vectors of length H_{kv}(which matches standard multi-head and GQA scaling up to small constant factors). In aggregate, this behaves like a streamed matrix–vector operation over an $S \times H_{kv}$ key matrix, giving:

$$F_{\text{score}} = 2SH_{kv}$$

The factor 2 reflects the standard GEMV convention (each multiply–accumulate counts as 2 FLOPs).

Value application (Attn · V)

After applying softmax to the scores, the attention weights are used to compute a weighted sum over the cached values. This again involves an effective $S \times H_{kv}$ matrix–vector style operation:

$$F_{\text{value}} = 2SH_{kv}$$

Total KV attention FLOPs

Combining the two:

$$F_{\text{attn,KV}} = F_{\text{score}} + F_{\text{value}} = 4SH_{kv}$$

This term captures the **sequence-length-dependent** attention cost during decoding.

3.3 FFN FLOPs (Unified Dense + MoE)

To match the parameter definitions in Section , we express FFN FLOPs using a **unified formulation** that works for both dense FFN layers and MoE layers.

Dense FFN FLOPs

A dense FFN consists of:

- an expansion GEMV: $H \to I$, and
- a contraction GEMV: $I \to H$.

Each GEMV costs $2HD$ FLOPs, giving:

$$F_{\text{ffn,dense}} = 4HI_{dense}$$

Dense FFN layers always have $EP = 1$.

MoE FFN FLOPs

For MoE layers:

- the **router** is applied to the full hidden vector:

  $$F_{\text{router}} = 2HN_{\text{exp}}$$

  where N_{exp} is the total number of experts.

- for each of the k selected experts for this token, the FFN computation is:

  $$F_{\text{expert}} = 4HI_{\text{moe}}$$

Thus the MoE FFN FLOPs for one token in one layer are:

$$F_{\text{ffn,moe}} = F_{\text{router}} + k \cdot F_{\text{expert}} = 2HN_{\text{exp}} + k(4HI_{\text{moe}})$$

Unified FFN FLOP Term

We now define **effective FFN FLOP parameters**:

- Effective FFN "matrix" dimension:

$$I_{\text{eff}} = \begin{cases} I_{\text{dense}}, & \text{dense layer}, \\ kI_{\text{moe}}, & \text{MoE layer}, \end{cases}$$

- Effective router multiplicity:

$$N_{\text{eff}} = \begin{cases} 0, & \text{dense layer}, \\ N_{\text{exp}}, & \text{MoE layer}. \end{cases}$$

With these definitions, both dense and MoE FFN FLOPs can be written in a **single unified form**:

$$F_{\text{ffn}} = 4HI_{\text{eff}} + 2HN_{\text{eff}}$$

This matches:

- Dense layer: $F_{\text{ffn}} = 4HI_{\text{dense}} + 2H \cdot 0 = 4HI_{\text{dense}}$
- MoE layer: $F_{\text{ffn}} = 4H(kI_{\text{moe}}) + 2HN_{\text{exp}} = 4HI_{\text{moe}}k + 2HN_{\text{exp}}$

3.4 LayerNorm and Elementwise FLOPs

LayerNorm, RMSNorm, residual additions, and small elementwise operations all scale linearly with H. Although these costs are much smaller than attention and FFN FLOPs, we include them for completeness.

We write:

$$F_{\text{norm}} = c_{\text{norm}}H$$

where c_{norm} is a small implementation-dependent constant (typically 5–20).

In practice, F_{norm} is much smaller than the projection, attention, and FFN FLOPs. In the per-device FLOP expressions below, we will neglect F_{norm} and write approximate equalities ($\approx$) to avoid clutter, noting that this introduces only a small relative error.

3.5 Per-Device FLOPs per Layer Under TP, SP, EP, and PP

For a single decoding token, the FLOPs for one transformer layer are:

$$F_{\text{layer}} = F_{\text{proj}} + F_{\text{attn,KV}} + F_{\text{ffn}} + F_{\text{norm}},$$

where:

- $F_{\text{proj}} = 2H^2 + 6HH_{kv}$ (Section 3.1),
- $F_{\text{attn,KV}} = 4SH_{kv}$ (Section 3.2),
- F_{ffn} is dense or MoE (Section 3.3),
- $F_{\text{norm}} = c_{\text{norm}}H$ (Section 3.4).

To find **per-device FLOPs**, we apply sharding from TP, SP, EP and then multiply by the number of layers on the PP stage.

TP (Tensor Parallelism)

TP shards all **GEMV/GEMM-like** FLOPs:

- projections (F_{proj}),

- attention score/value FLOPs ($F_{\text{attn,KV}}$),
- FFN GEMMs (dense or MoE experts).

Thus:

$$F_{\text{tensor}}^{\text{device}} = \frac{1}{TP} F_{\text{tensor}}$$

Normalization FLOPs F_{norm} are tiny and unsharded:

$$F_{\text{norm}}^{\text{device}} = F_{\text{norm}}$$

SP (Sequence Parallelism)

SP shards the **sequence dimension** across SP ranks.
Thus **only** the sequence-dependent FLOPs:

$$F_{\text{attn,KV}} = 4SH_{kv}$$

are reduced:

$$F_{\text{attn,KV}}^{\text{device}} = \frac{1}{TP \cdot SP}(4SH_{kv})$$

SP does **not** reduce:

- F_{proj} (single token),
- F_{ffn} (dense or MoE),
- router FLOPs,
- normalization FLOPs.

EP (Expert Parallelism)

EP applies only to MoE layers:

- Router FLOPs use the full hidden state and are **not sharded**.
- Expert FFN GEMMs are sharded across EP:

$$F_{\text{expert}}^{\text{device}} = \frac{k}{EP}\,(4HI_{\text{moe}})$$

and may also be TP-sharded if FFN GEMMs follow the same path:

- If experts use TP GEMMs: divide by TP again.
- If experts are **only EP-sharded**, do **not** include $1/TP$.

Dense layers always have $EP = 1$.

PP (Pipeline Parallelism)

PP assigns whole layers to stages.
Each device in a PP stage owns:

$$\frac{L}{PP}\ \text{layers.}$$

Thus:

$$F_{\text{token, device}} = \frac{L}{PP}\left(F_{\text{proj}}^{\text{device}} + F_{\text{attn,KV}}^{\text{device}} + F_{\text{ffn}}^{\text{device}} + F_{\text{norm}}\right)$$

Total Per-device FLOPs:

Droping the negligible F_{norm} and also substituting everything yields the **final fully expanded expression** per-device FLOPs for a single decoded token:

$$\boxed{F_{\text{token,device}} \approx \frac{L}{PP}\left(\frac{2H^2 + 6HH_{kv}}{TP} + \frac{4HI_{\text{eff}}}{TP \cdot EP} + \frac{4SH_{kv}}{TP \cdot SP} + 2HN_{\text{eff}}\right)}$$

For a **dense MLP model**: $I_{\text{eff}} = I_{\text{dense}}, \quad N_{\text{eff}} = 0, \quad EP = 1$

For a **MoE model**: $I_{\text{eff}} = kI_{\text{moe}}, \quad N_{\text{eff}} = N_{\text{exp}}, \quad EP \geq 1$

3.6 Prefill FLOPs (Supplementary Reference)

The expressions in Sections 3.1–3.5 apply strictly to **decoding**, where only one new token is processed per step and FLOPs scale as $O(S)$ through the $F_{\text{attn,KV}}$ term.

For completeness, we summarize the **prefill (full-sequence)** FLOPs here. Prefill is **GEMM-dominant** and substantially more expensive than decoding, but its cost is incurred **once per request**, not per generated token.

A coarse scaling for prefill FLOPs is:

$$F_{\text{prefill}} = O(BSLH^2) \; + \; O(BLS^2H_{kv})$$

- The first term corresponds to projections and FFN blocks across all S input tokens.
- The second term corresponds to full attention score computation over the $S \times S$ attention matrix per layer.

Why this section is supplementary:
Prefill FLOPs do **not** affect decoding throughput (TPS_{single}) and do not enter the per-device roofline model (Sections 5.5, 7). They only contribute to **TTFT** (time-to-first-token), which is modeled in Section 9.

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4. Compute vs. Memory Bound (Roofline Model)

Sections 2 and 3 derived:

- the **per-token memory traffic** ($T^{eff}_{\text{token,device}}$), including FlashAttention effects.
- the **per-token FLOPs per device** on this PP stage ($F_{\text{token,device}}$), and

We now combine them using a standard roofline model to determine the **local per-token latency**.

Let:

- R_{GPU}: sustained device compute throughput (FLOPs/s),
- $B_{\text{eff,mem}}$: sustained device memory bandwidth (bytes/s).

Both reflect *sustained* performance, not peak specs.

4.1 Operational Intensity (Ops:Byte)

The **operational intensity** for decoding on this device is:

$$\text{OI} = \frac{F_{\text{token,device}}}{T^{eff}_{\text{token,device}}} \quad (\text{FLOPs per byte})$$

High-level interpretation:

- **High OI** → more FLOPs per byte → *compute-bound*
- **Low OI** → fewer FLOPs per byte → *memory-bound*

This rating is compared to the device's memory-to-compute ratio:

$$\frac{R_{\text{GPU}}}{B_{\text{eff,mem}}}$$

If:

$$\text{OI} > \frac{R_{\text{GPU}}}{B_{\text{eff,mem}}} \quad \Rightarrow \quad \text{compute-bound}$$

else:

$$\text{OI} < \frac{R_{\text{GPU}}}{B_{\text{eff,mem}}} \quad \Rightarrow \quad \text{memory-bound}$$

Dominant-Term Approximation

In practice, the OI is often approximated using only the largest FLOP and traffic terms:

- FLOPs dominated by:

$$\max\big(2H^2,\ 4HI_{\text{eff}},\ 4SH_{kv}/SP\big)$$

- Memory traffic dominated by the KV term:

$$\frac{2SH_{kv}}{TP \cdot SP}\, b$$

Thus for long-context decoding:

$$\mathrm{OI} \approx \frac{4SH_{kv}/(TP \cdot SP)}{2SH_{kv}/(TP \cdot SP)\ b} = \frac{2}{b}$$

which explains why long-context dense models almost always appear **memory-bound**.

4.2 Compute-Bound Time

Given the per-token FLOPs on this device (Section 3.5):

$$t_{\text{compute}} = \frac{F_{\text{token,device}}}{R_{\text{GPU}}}$$

This is the time assuming unlimited memory bandwidth.

4.3 Memory-Bound Time

Given the effective memory traffic per token (Section 2.4):

$$t_{\text{mem}} = \frac{T^{eff}_{\text{token,device}}}{B_{\text{eff,mem}}}$$

This is the time assuming compute is free.

For long-context LLMs, this term is often dominant due to KV-cache reads.

4.4 Local Device Time Bound

The per-token latency for this PP stage is:

$$t_{\text{local}} = \max\left(t_{\text{compute}},\ t_{\text{mem}}\right)$$

This is the roofline principle:

> **Decoding is limited by whichever is slower — compute throughput or memory bandwidth.**

---

5. Communication Time During Decoding

Section 4 defined the **local per-token latency** on each device:

$$t_{\text{local}} = \max(t_{\text{compute}}, t_{\text{mem}})$$

We now incorporate the **inter-device communication time** that arises during decoding under distributed parallelism. In the nested parallelism structure used throughout this document:

$$\mathbf{PP} \;\rightarrow\; \mathbf{EP} \;\rightarrow\; \mathbf{TP} \;\rightarrow\; \mathbf{SP}$$

each axis contributes its own per-token communication term. All communication costs follow the standard α–B model:

$$t_{\text{comm}} = \alpha + \frac{\text{message size}}{B_{\text{eff}}}$$

where α is the collective or hop latency, and B_{eff} is the sustained bandwidth of the communication path.

The parameters α and B_{eff} in this model are not abstract: they are **topology-dependent physical properties** of the underlying interconnect. Different parallelism domains—TP, EP, SP, and PP—may be mapped to **different network fabrics** or different portions of the same physical topology (e.g., NVSwitch star within a node, 2D/3D torus across nodes, or hybrid switch-plus-fabric designs). Consequently, each collective type sees its own communication characteristics, with potentially different latency constants and effective bandwidths. To keep the analysis general, we denote these as α_{XP} and $B_{\text{eff},XP}$ for TP, EP, SP, and PP respectively. Their actual numerical values depend on the system's physical layout, routing scheme, and bisection bandwidth properties (e.g., constant-hop NVSwitch vs. hop-scaling torus fabrics). The following sections therefore use α_{XP} and $B_{\text{eff},XP}$ as **collective-specific, topology-aware** parameters, to be instantiated according to the actual deployment mapping.

Message sizes and their shard structure

To remain consistent with the compute and memory models in previous sections:

- **PP** communication uses **activation shards** of width $\approx H/TP$
  (rank-aligned, TP-preserving pipeline mapping)
- **EP** communication uses **full activations** for each expert, with a message size of $k \cdot H$
  (MoE routing and expert FFNs require full-hidden-state inputs)
- **TP** communication uses **activation shards** of width $\approx H/TP$
- **SP** communication uses **KV-cache shards** of size $\frac{S}{SP} \cdot \frac{2H_{kv}}{TP}$

These message sizes match the per-device shard dimensions used in Section 2.

5.1 Pipeline Parallel (PP) Hop

Pipeline Parallelism (PP) forwards activations from one pipeline stage to the next. Because TP is nested inside PP, high-performance implementations (e.g., Megatron-LM PP, DeepSpeed PP, NVIDIA NeMo, and FasterTransformer) preserve the **TP rank alignment** across all PP stages. That is, TP rank i in stage s corresponds directly to TP rank i in stage $s{+}1$.

This alignment has an important consequence:
each device only needs to forward its own TP shard of the hidden state, not a full H-dimensional vector. The full activation is conceptually transferred across the PP boundary, but it is split naturally across TP separate device-to-device links.

Thus the PP hop behaves as a **single, shard-sized point-to-point transfer**, with message size $\approx H/TP$ per device.

For a single token, the latency of this hop is modeled as:

$$t_{PP} = \alpha_{PP} + \frac{(H/TP)\,b}{B_{\text{eff,PP}}}$$

This shard-preserving PP design avoids the extra TP collectives that would be required if stages exchanged full activations and then re-sharded them. Maintaining TP rank consistency across stages therefore yields a significantly faster pipeline, and is the standard strategy in modern LLM training and inference systems.

5.2 Expert Parallel (EP) All-to-All (MoE Dispatch and Combine)

MoE layers require exchanging token activations across the expert-parallel (EP) dimension. In contrast to TP collectives—which must perform a two-phase reduce-then-broadcast—EP communication follows a **single-pass dispatch-and-combine pattern**:

1. Token activations are **dispatched** to the appropriate experts (top-k routing).
2. Expert outputs are **returned** to the originating ranks within the same all-to-all schedule.
3. Expert FFN computation occurs locally; no second communication phase is needed.

Because dispatch and combine are fused into a single all-to-all exchange, EP communication behaves as a **one-directional exchange**, not a two-pass reduction. This leads to noticeably different latency characteristics compared to TP and SP collectives.

Within a single DP replica, the nested structure is PP → EP → TP → SP:

- EP groups **enclose** TP groups.
- MoE routing consumes the **full hidden-state vector** $h \in \mathbb{R}^H$.
- Expert FFNs also require full-H inputs before internal TP sharding.

Thus, EP communication always operates on **full-H activations**.

Let k denote the number of active experts per token (top-k routing).
Each device transmits approximately kHb bytes per token during routing.

5.2.1 Ring-Style EP All-to-All

A ring all-to-all across EP devices performs $(EP-1)$ exchange rounds.
A **per-token, per-layer** latency model is:

$$t_{EP}^{\text{ring}} = (EP-1)\alpha_{EP} + (EP-1)\frac{kH\,b}{EP \cdot B_{\text{eff,EP}}}$$

Interpretation:

- Communication depth grows linearly with $(EP-1)$.

- Each device sends the equivalent of kHb bytes per token, amortized across the rounds.
- This provides a conservative (higher-latency) bound.

5.2.2 Log-Style (Tree/Recursive) EP All-to-All

Optimized implementations can reduce collective depth via recursive-doubling or tree-style schedules, and the **per-token, per-layer** latency can be modeled as:

$$t_{EP}^{\text{tree}} \approx \lceil \log_2(EP) \rceil \alpha_{EP} + \frac{kH\,b}{B_{\text{eff,EP}}}$$

Here:

- The exchange requires only $\lceil \log_2(EP) \rceil$ rounds.
- The per-device payload remains kHb; only scheduling depth changes.

For dense models ($EP = 1$), we have $t_{EP} = 0$.

5.3 Tensor Parallel (TP) Communication

TP groups compute each layer in parallel across TP devices. Each device holds a shard of width $\approx H/TP$ and must participate in synchronization collectives per layer. These collectives must both **combine** partial results and **redistribute** the combined outputs, which is why TP collectives typically have a factor of 2 (reduce + broadcast).

5.3.1 TP Ring All-Reduce / All-Gather

For a ring all-reduce, the **per-token, per-layer** latency can be modeled as:

$$t_{TP}^{\text{ring}} = 2(TP-1)\alpha_{TP} + 2\frac{TP-1}{TP} \cdot \frac{(H/TP)\,b}{B_{\text{eff,TP}}}$$

Interpretation of the factor of 2:

- A full all-reduce consists of **two phases**:
  reduce-scatter (devices exchange and accumulate partial results) and
  all-gather (devices exchange and reconstruct the full output).
- Each phase sends a fraction $\frac{TP-1}{TP}$ of the local shard.
- This two-phase structure is inherent to all-reduce semantics and cannot be avoided when combining TP-sharded partial results.

Thus, TP ring all-reduce inherently introduces a **two-pass communication pattern**, unlike EP all-to-all (single pass) or PP hops (single peer exchange).

5.3.2 TP Tree All-Reduce / All-Gather

For a tree algorithm, the **per-token, per-layer** latency can be modeled as:

$$t_{TP}^{\text{tree}} \approx 2\lceil \log_2(TP) \rceil \alpha_{TP} + 2\frac{(H/TP)\,b}{B_{\text{eff,TP}}}$$

Interpretation:

- The **tree reduction** accumulates partial results across a tree of depth $\lceil \log_2(TP) \rceil$.
- The **tree broadcast** redistributes the final output.
- Both stages transfer the same $(H/TP)\, b$ shard, hence the factor of 2.

Tree-based all-reduce reduces latency relative to ring, especially when:

- TP is large,
- the interconnect latency α_{TP} dominates,
- or the shard size (H/TP) is relatively small.

5.4 Sequence Parallel (SP) Communication

Sequence Parallelism (SP) partitions the KV cache along the sequence dimension. For each Transformer layer, the layer's KV cache has shape $S \times \frac{2H_{kv}}{TP}$, and SP assigns each rank a contiguous shard of this cache of size $(S/SP) \cdot \frac{2H_{kv}}{TP}$. Because every layer maintains its own K and V representations, each layer independently uses its own sharded KV matrix during decoding.

When a new token arrives, its self-attention in a given layer must incorporate contributions from all S past positions. Under SP, each rank holds only a portion of these positions, so each layer performs a "scatter–gather" style exchange over its KV shard: each rank first computes partial attention results using the tokens in its (S/SP) slice and then exchanges these partial results with all other SP ranks to assemble the full attention output for that layer. This two-pass "scatter–gather" structure (distributing partial contributions and then collecting the aggregated results) leads to a per-layer communication cost with effective message size proportional to $(S/SP) \cdot \frac{2H_{kv}}{TP}$ and is why SP communication inherently has the **same factor of 2** seen in reduce–broadcast collectives. Unlike EP (which is a single-pass dispatch-and-combine), SP attention depends on *both* sending out partial results and receiving all other ranks' partial results.

While tree-based variants are theoretically possible, practical implementations (e.g., Megatron-LM SP, DeepSpeed-Ulysses, and fused attention kernels) consistently use ring-style schedules: KV shards are large, non-contiguous in memory, and must be processed in left-to-right sequence order, making tree-style SP impractical.

Thus, for modeling purposes, we assume **ring-style SP communication**.

SP Ring Communication Latency

A ring with SP ranks performs $(SP-1)$ exchange rounds, and the scatter–gather structure requires two full passes, giving the factor of 2. The **per-token, per-layer** latency can therefore be expressed as:

$$t_{SP} = 2(SP-1)\alpha_{SP} + 2\frac{SP-1}{SP} \cdot \frac{\left(\frac{S}{SP} \cdot \frac{2H_{kv}}{TP}\right) b}{B_{\text{eff,SP}}}$$

Interpretation:

- The first pass distributes (scatters) partial attention contributions.
- The second pass collects (gathers) the contributions needed to form final attention outputs.
- Both passes operate on KV shards of size $\frac{S}{SP} \cdot \frac{2H_{kv}}{TP}$.

This two-pass requirement distinguishes SP from the **single-pass** EP all-to-all and from the simple one-hop PP communication. SP behaves more like TP all-reduce in its communication structure, but the underlying data patterns (KV shards and attention sequencing) make ring-style scheduling the dominant practical choice.

5.5 Total Communication Time Per Token on a PP Stage

Sections 5.1–5.4 provide **per-token, per-layer** communication costs for each parallelism axis (TP, EP, SP), and a **per-token, per-hop** cost for PP.
We now clarify how these terms combine to form the total per-token communication time on a given pipeline-parallel (PP) stage.

Per-layer vs. per-stage normalization

A Transformer layer contains exactly one Attention block and one MLP/MoE block. Each block triggers a fixed number of communication collectives, and within each layer, TP, EP, and SP collectives are stricly ordered:

- **Attention**
  - 1 TP collective
  - 1 SP collective (if SP is enabled)
- **MLP (dense)**
  - 1 TP collective
- **MoE block**
  - 1 EP all-to-all
  - 1 TP collective (for expert-FFN output projection)

These collectives must complete before the token can advance to the next layer. Since a PP stage contains L/PP *sequential* layers, the total communication work for that stage is:

- n_{TP} TP collectives per layer
- n_{EP} EP collectives per layer (0 for dense layers, 1 for MoE layers)
- 1 SP collective per layer (if SP is enabled)

Because TP, EP, and SP operations within each layer depend on one another (e.g., TP → SP in attention and EP → TP in MoE), they are **strictly sequential** and do not overlap. Thus, the **per-token communication time accumulated over the entire PP stage** is:

$$t_{\text{comm,stage}} = \frac{L}{PP}\left(n_{TP}\, t_{TP} + n_{EP}\, t_{EP} + t_{SP}\right)$$

Where:

- t_{EP}, t_{TP}, t_{SP}
  are the **per-token, per-layer** communication costs given in Sections 5.2–5.4.
- n_{TP}, n_{EP}
  are the counts of TP and EP collectives executed inside each layer on this PP stage. Typical values are:
  - Dense layers: $n_{TP} = 2$, $n_{EP} = 0$
  - MoE layers: $n_{TP} = 2$, $n_{EP} = 1$

Adding PP hop cost

The PP hop is different: it is a **per-token, per-hop** cost rather than a per-layer cost. A token is forwarded once from PP stage s to stage $s+1$, with latency t_{PP} as defined in Section 5.1.

Thus, the total per-token communication time for this stage is:

$$\boxed{t_{\text{comm}} = \frac{L}{PP}\left(n_{TP}\, t_{TP} + n_{EP}\, t_{EP} + t_{SP}\right) + t_{PP}}$$

Interpretation

- The first term accumulates **all TP, EP, and SP collectives** required by the L/PP layers on this PP stage.
- The second term accounts for the **one PP hop** that forwards the token to the next stage.
- This combined expression represents the **total communication work** per token for the stage. Whether this communication becomes the latency bottleneck or is hidden by overlap is addressed in Section 4's roofline-style model and Section 6's end-to-end pipeline analysis.

Summary of Collective Types and Message Sizes

Parallelism	Occurs in	Collective Type	Passes	Message Size (per device)
PP	between layers	point-to-point	1	$(H/TP)\, b$
TP	attn + FFN	all-reduce (ring/tree)	2	$(H/TP)\, b$
EP	MoE FFN	all-to-all	1	$kH\, b$
SP	attention	scatter–gather	2	$(S/SP) \cdot (2H_{kv}/TP)\, b$

This table highlights that EP is full-H and one-pass,
TP and SP are shard-sized and two-pass,
and PP forwards only TP-aligned shards.

Practical Guidance: When to Use Ring vs. Tree Collectives

The choice between ring-style and tree-style collective algorithms depends strongly on the **physical interconnect topology**:

- **NVSwitch / fully-connected crossbars (constant-hop latency):**
  Ring collectives often provide higher effective bandwidth because all links can operate concurrently with no hop-distance penalty.

- **2D/3D torus, dragonfly, and multi-hop switch fabrics:**
  Tree or recursive-doubling collectives typically reduce depth and latency, since ring communication cost scales with hop count. Low-latency links benefit more from tree scheduling.

- **Sequence Parallelism (SP):**
  SP operations must traverse KV shards in strict left-to-right order. Tree-style SP is rarely implemented in practice; **ring is the standard**.

This guidance explains why Section 5 provides both **ring** and **tree** expressions for EP and TP, but uses **ring** exclusively for SP.

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6. End-to-End Latency, Throughput, and Partition Strategy

This section integrates memory limits, compute, communication, and pipeline behavior into a complete end-to-end decoding performance model. We discuss:

1. Feasible model partitioning via **HBM limits**
2. Local per-token latency via the **compute–memory roofline**
3. Per-token communication latency from **TP/EP/SP/PP**
4. Overlap strategies between compute and communication
5. Throughput (**TPS**, **TTPS**) based on PP-stage bottlenecks
6. **TTFT**, including prefill on a separate cluster and PP-stage latency summation

---

6.1 Model Partition Strategy from HBM Constraints

A parallel configuration (DP, PP, EP, TP, SP) is **feasible** only if each device can store:

- its **parameter shard**,
- its **KV-cache shard**, and
- the **activation workspace** needed for a single decoding token,

within the available HBM capacity M_{HBM}.

We define the total per-device static footprint as

$$M_{\text{device}}^{\text{total}} = M_{\theta,\text{device}} + M_{\text{act,device}} + M_{\text{KV,device}} \;\leq\; M_{\text{HBM}}$$

Using the per-device memory expressions derived in Section 1, the **fully expanded** form is:

$$\frac{L}{PP}\left(\frac{H^2 + 3HH_{kv}}{TP} \;+\; \frac{2HIN_{\text{exp}}}{TP \cdot EP} + \;(4H + 2H_{kv}) + \frac{2SH_{kv}}{TP \cdot SP}\right) b + \frac{VH}{TP} b \leq \;M_{\text{HBM}}$$

where:

- the bracketed term is the **intermediate PP-stage** footprint (parameters, activations, KV cache),
- and the final $\frac{VH}{TP} b$ term models the **worst-case embedding / LM-head overhead** on boundary PP stages
  (for both tied and untied heads, one of W_{emb} or W_{lm} of size VH resides on a given device, TP-sharded but not PP-sharded).
- for a **dense MLP model**: $I = I_{\text{dense}}$, and $N_{\text{exp}} = EP = 1$.
- for a **MoE model**: $I = I_{\text{moe}}$, with $N_{\text{exp}} > 1$ and $EP \geq 1$.

6.2 Local and Networking Per-Token Latency

Compute-bound latency

$$t_{\text{compute}} = \frac{F_{\text{token,device}}}{R_{\text{GPU}}}$$

Memory-bandwidth-bound latency

$$t_{\text{mem}} = \frac{T^{\text{eff}}_{\text{token,device}}}{B_{\text{eff,mem}}}$$

Roofline local latency

$$t_{\text{local}} = \max(t_{\text{compute}},\ t_{\text{mem}})$$

Collective communication latency

$$t_{\text{comm}} \approx \frac{L}{PP}\big(n_{TP}\, t_{TP} + n_{EP}\, t_{EP} + t_{SP}\big) + t_{PP}$$

Overlap Models

No overlap (conservative)

$$t^{\text{no-overlap}}_{\text{token}} = t_{\text{local}} + t_{\text{comm}}$$

Full overlap (optimistic)

$$t^{\text{overlap}}_{\text{token}} = \max(t_{\text{local}},\ t_{\text{comm}})$$

Partial overlap (empirical)

$$t_{\text{token}} \approx \max(t_{\text{local}},\ \rho\, t_{\text{comm}})\,, \quad \rho \in [0.3, 1]$$

where:

- For TP collectives, NVIDIA NeMo reports substantial overlap via asynchronous all-gather and reduce-scatter operations [NeMo User Guide 2025].
- For MoE EP all-to-all collectives, recent work demonstrates only partial overlap due to token routing and expert locality constraints [MegaScale-MoE 2025].
- Attention-related communication (SP) and pipeline transfers exhibit limited overlap because of strict data dependencies [FlashAttention 2022; PipeDream 2019].
- These observations motivate modeling overlap through a factor $\rho \in [0.3, 1]$

6.3 TPS and TTPS — Pipeline Throughput

Recall from Section 6.2 that for a given PP stage we model the **per-token, per-stage** latency as

$$t_{\text{token}} \approx \max\big(t_{\text{local}},\ \rho\, t_{\text{comm}}\big)$$

where t_{local} is the local compute/memory time, t_{comm} is the communication time for that stage, and $\rho \in [0.3, 1]$ models communication–compute overlap.

For pipeline stage j, we write this explicitly as

$$t_{\text{stage},j} = \max\big(t_{\text{local},j},\ \rho\, t_{\text{comm},j}\big)$$

During steady-state decoding, different PP stages process **different tokens** concurrently. The overall throughput of a single DP replica is therefore limited by the **slowest** pipeline stage:

$$TPS_{\text{single}} \approx \frac{1}{\max_j t_{\text{stage},j}}$$

Across DP fully independent replicas (no cross-replica coupling), the total cluster throughput scales linearly:

$$TTPS \approx DP \cdot TPS_{\text{single}} = DP \cdot \frac{1}{\max_j t_{\text{stage},j}}$$

Here:

- The **throughput bottleneck** is $\max_j t_{\text{stage},j}$ (the slowest stage).
- The **pipeline depth / first-token traversal time** is $\sum_{j=1}^{PP} t_{\text{stage},j}$, which appears in the TTFT expression in Section 6.4.

6.4 TTFT — Time To First Token (Prefill on a Separate Cluster)

This section models TTFT assuming the **prefill runs on a separate cluster**, and the decoding cluster receives the final KV cache before emitting the first token.

For clarity, TTFT consists of three sequential phases:

1. **Prefill latency** (could be on a separate cluster)
2. **KV-cache transfer**
3. **Pipeline traversal delay** (empty pipeline warm-up)

Only after all PP stages have processed the first token can the first decoded token be emitted.

Prefill latency (other cluster)

Let the prefill cluster have:

- Compute rate: $R_{\text{GPU,pre}}$
- Memory bandwidth: $B_{\text{eff,mem,pre}}$

Then:

$$t_{\text{prefill}} \approx \max\left(\frac{F_{\text{prefill}}}{R_{\text{GPU,pre}}}, \frac{T_{\text{prefill}}}{B_{\text{eff,mem,pre}}}\right)$$

KV-cache transfer

Total KV size:

$$M_{\text{KV,total}} = 2SH_{kv}b \cdot L$$

If inter-cluster link bandwidth is $B_{\text{link,cluster}}$:

$$t_{\text{KV-transfer}} \approx \frac{M_{\text{KV,total}}}{B_{\text{link,cluster}}}$$

Pipeline traversal latency (decoding cluster)

The first decoded token must traverse all PP stages **sequentially**, since the pipeline is empty:

$$t_{\text{decode,first-token}} \approx \sum_{j=1}^{PP} t_{\text{stage},j}$$

Final TTFT

$$TTFT \approx t_{\text{prefill}} + t_{\text{KV-transfer}} + \sum_{j=1}^{PP} t_{\text{stage},j} + t_{\text{startup}}$$